

BATTERY PASSPORT - RESEARCH PROTOTYPE

Not a certified battery passport. Fields are oriented on Regulation (EU) 2023/1542, Annex XIII.

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1. Battery and data-source provenance

Battery ID: B0005

Data source: NASA

Provenance: NASA public dataset, processed by this project

Model: base XGBoost-fusion model (validated routing) (encoder: old_v1)

2. State of health

State of health: 71.7 %

90% interval: 66.8 - 76.6 %

Interval method: split conformal, 90% target ($\alpha = 0.1$): one fixed half-width calibrated on held-out batteries of the training pool and applied to every battery; it is an in-domain guarantee only and is not adaptive. Measured capacity checkpoints used: 0.

90% conformal interval calibrated on known sources

3. Expected remaining life

Expected remaining life: not shown

Reason: not available for this battery

RUL is shown only when the nearest known source is NASA or MIT and the battery is not flagged

4. Cycle count

Cycles available for this battery in this app: 168 Assessed cycle: 168

5. Trust status of this assessment

No distribution shift detected (this is not a guarantee)

Nearest known source: NASA score nll_min = -16.46 (flagged if ≥ -5.52)

Stated detection rate: the check flagged 81.9% of batteries from a source it had never seen and wrongly flagged 11.0% of batteries from known sources (leave-one-source-out over 16 sources). It is weakest for mich, NASA, snl. About 1 in 5 unfamiliar batteries are not detected; check against a measured capacity when possible.

Typical SOH error (MAE, points): 3.2 for the nearest known source vs about 5.9 on an unseen source

Guarantee: none - this is a screening check.

Input-sanity check: not run: built-in dataset (the one-class SVM check is applied to uploads)

6. Disclaimer

RESEARCH PROTOTYPE - NOT A CERTIFIED BATTERY PASSPORT. This document is generated by a university research project (CellSense). It is not issued under, and makes no claim of compliance with, Regulation (EU) 2023/1542. The numbers are model estimates with the stated uncertainty and limits; they are not measurements and must not be used for safety-critical, commercial or legal decisions without an independent measured capacity test.

Routing was selected on held-out data, so it is a deployment choice, not an unbiased test result.